

Open Source 101

An Introduction to Open Source

Understanding open source fundamentals

aws.amazon.com/opensource

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SECTION 01

What Is Open Source?

Understanding the foundation of collaborative software development

Defining Open Source Software

Open source software is software whose source code is made available to the public. Anyone can inspect, modify, and distribute it.



Open Source Code

The code is publicly available for anyone to read and study.



Freedom to Modify

Users can change, improve, and adapt the software freely.



Freedom to Distribute

Modified versions can be shared with the world.



Community Driven

Built collaboratively by contributors worldwide.

Why Open Source Matters

Open source software can be freely viewed, used, modified, and improved by anyone. That is not a technical detail. It is a decision about who gets to participate in building the technology the world depends on.

Built to be open

Open source software is released under licenses that allow anyone to freely view, use, modify, and improve it.

Shared infrastructure, shared improvement

Everyone builds on the same base and every improvement flows back to everyone. Linux, Kubernetes, Python, and PostgreSQL improve because the community uses and contribute to them collectively.

Transparency creates accountability

Open source can be inspected by anyone, so bugs and security flaws are discovered faster, claims can be verified, and trust is earned rather than asked for.

Problems get solved once, not repeatedly

Every hard problem solved in the open becomes inherited knowledge. Databases, web servers, ML frameworks, and cryptography libraries were each solved once, shared openly, and now underpin almost everything built since.

Forks keep power distributed

When critical infrastructure is proprietary, the owner sets the terms for everyone who depends on it. Open source means no company can raise prices arbitrarily, change the rules, or discontinue something the world relies on.

Open Source is the Core

The internet runs largely on open source. Cloud computing is built on open source foundations. Modern AI research and tooling depend on open source frameworks. Open source is not an alternative to mainstream technology. It is the foundation mainstream technology is built on.

Why Companies Invest in Open Source

01

Faster innovation through shared R&D

Instead of building alone, developers across organizations contribute improvements. Bugs get fixed faster and features evolve quicker than any closed internal team could manage alone. Open source becomes shared research and development.

02

Becoming the industry standard

When your technology becomes the open standard, developers learn your ecosystem, tools integrate with your platform, and your organization becomes central to where innovation happens. Google released Kubernetes and the industry standardized on it.

03

Real work, visible to everyone

Open source is a transparent hiring pipeline. Companies can evaluate real commits, real code reviews, and real collaboration rather than relying on credentials alone.

04

Reducing development cost

Instead of reinventing components, companies reuse community software that are maintained by a global community and share maintenance costs globally. So the development and upkeep costs are shared rather than on one organization, everyone benefits.

05

Building trust through transparency

Open source software can be independently audited, inspected, and verified by anyone. The code is visible, the license is public, and security claims can be checked rather than taken on faith.

06

Ecosystem growth drives business growth

Open source software grows adoption. Adoption creates a user base. That base generates demand for cloud hosting, enterprise support, managed services, and consulting. The software is free. The value built around it is not.

The Core Principles Behind Open Source

aws.amazon.com/what-is/open-source

01

Community

An open-source community is a group of people who come together to achieve a common purpose. Their shared values and goals guide decision making and drive the project forward.

02

Transparency

Open source projects ensure that everyone has access to the information necessary for doing their best work. With full visibility, team members can make decisions and build on each other's discoveries.

03

Governance

Community projects encourage teamwork, so a group solves problems that individuals alone cannot. For example, members in any open source initiative contribute changes to improve what someone else has built. You can also establish rules so others can modify the solution as needed in the future.

04

Collaboration

When a lot of people try to solve the same problem, the ideas that work best for everyone rise to the top. Open-source makes that happen by letting anyone jump in and improve the code

Open Source vs. Closed Source

OPEN SOURCE

- Source code is publicly visible
- Anyone can contribute improvements
- Community-driven development
- Examples: Linux, Firefox, Python, Git
- Transparent, you can audit it
- Maintained by global contributors

VS

CLOSED SOURCE

- Source code is kept private
- Only the company modifies it
- Controlled by the owning organization
- Examples: Windows, Adobe Photoshop
- Usually requires paid licensing
- Maintained by internal staff only

Notable Open Source Projects



Firefox

Web Browser

Mozilla's open source browser started the modern browser wars. It pioneered extensions, privacy tools, and web standards. Trusted by hundreds of millions globally.



Android

Mobile OS

Google's Android is built on the Linux kernel and open source components. The AOSP (Android Open Source Project) lets anyone build their own Android-based OS.



Python

Programming Language

One of the world's most popular programming languages. Python's open source nature and rich ecosystem of libraries make it the default choice for AI, data science, and web.



Docker

Containerization

Docker revolutionized software deployment by packaging apps into containers. Built on open source foundations, it became the industry standard almost overnight.



VS Code

Code Editor

Microsoft's Visual Studio Code is the world's most popular code editor. It's fully open source on GitHub, with thousands of community-built extensions.



PostgreSQL

Database

The world's most advanced open source relational database. Trusted by companies like Apple, Instagram, and Netflix for mission-critical workloads requiring reliability.

SECTION 02

Open Source Foundations

What Is an Open Source Foundation and Why Does It Matter?

Open source software can be written by anyone, but when it becomes critical infrastructure that organizations and communities rely on, it needs a legal home. A foundation is a nonprofit organization that holds the intellectual property, protects the trademarks, and ensures no single company can control or take ownership of something that belongs to everyone.

Neutral ownership no one can claim

Without a foundation, an open source project is effectively owned by whoever started it or funds it. A foundation takes legal ownership of the code, trademarks, and brand, holding them in trust for the entire community. No company can buy or acquire.

Safe Space for competitors to collaborate

Competing companies often benefit from a neutral third party when collaborating on shared infrastructure. A foundation provides that legal structure. Google, Microsoft, Amazon, and Meta can all contribute to the same project because the foundation, not any of them, owns it.

Governance and Continuity

Individual maintainers move on. Companies change priorities. A foundation creates governance structures, voting rights, decision processes, and codes of conduct that keep a project healthy and moving forward regardless of who comes and goes.

Contributing Without Risk

The foundation handles the intellectual property and licensing framework. This provides legal clarity and reduces risk. Without this structure, contributing code to a project can expose individuals and employers to legal uncertainty.

Governance and Fairness

Governance structures are designed to ensure that project direction is driven by technical merit and community consensus, not just by whichever company has the biggest team of contributors.

Built to last

A company can pivot, shut down, or get acquired. A nonprofit foundation cannot be bought. A foundation provides the governance structure that helps ensure it remains available and open for the long term.

The Linux Foundation

The Linux Foundation is one of the world's leading homes for collaboration on open source software, hardware, standards, and data. Its projects are critical to the world's infrastructure including Linux, Kubernetes, Node.js, PyTorch, and more.

What the Linux Foundation Does

Hosts open source projects in a neutral space

Companies that compete in the market can collaborate here without conflict of interest.

Provides governance and legal structure

Handles trademarks, IP, licensing, and community rules so contributors can focus on code.

Funds development and events

Training, certification, conferences (KubeCon, Open Source Summit) and developer grants.

Manages sub-foundations

Specialized foundations for cloud (CNCF), AI (LF AI & Data), agents (AAIF), fintech (FINOS), and more.

Major Projects & Sub-Foundations

Linux Kernel - Project

The Linux kernel powers the vast majority of today's servers, Android devices, and cloud platforms.

Kubernetes - CNCF Project

The standard for deploying and managing containerized applications at scale. Used by AWS, Google, Microsoft.

PyTorch - PyTorch Foundation

Facebook's open source AI framework - now the most widely used platform for deep learning research and production.

Node.js - OpenJS Foundation

The JavaScript runtime powering millions of web servers worldwide, from startups to Netflix.

The Linux Foundation - Specialised Homes for Every Domain

The Linux Foundation also hosts dedicated sub-foundations for specific fields.

As open source expanded into cloud, AI, security, data, and automotive, the Linux Foundation created specialised sub-foundations. Each is a neutral, vendor-governed home for projects.

Cloud Native Computing Foundation (CNCF)

Governs the cloud native ecosystem. Home to Kubernetes, Prometheus, Envoy, Helm, and Argo. If software runs in containers at scale, it likely depends on a CNCF project.

Notable projects:

Kubernetes, Prometheus, Helm, Argo, Fluentd

LF AI & Data Foundation

Dedicated to open source AI and machine learning. Hosts projects across the entire AI pipeline from data preparation to model deployment and evaluation.

Notable projects:

ONNX, Delta Lake, Horovod, Kedro, BeeAI, Docling

Agentic AI Foundation (AAIF)

Announced December 2025. Ensures agentic AI evolves as open, interoperable infrastructure. AWS, Google, Microsoft, Bloomberg, and Cloudflare are founding members.

Notable projects:

MCP (Anthropic), AGENTS.md (OpenAI), goose (Block)

OpenSSF (Open Source Security Foundation)

Strengthens the security of open source software across the entire supply chain. Responds to vulnerabilities, builds security tooling, and sets best practices.

Notable projects:

Sigstore, SLSA, Scorecard, Alpha-Omega, Allstar

Other notable sub-foundations: LF Edge, LF Energy, LF Networking, Academy Software Foundation, Confidential Computing Consortium, Open Mainframe Project, Hyperledger Foundation

The Apache Software Foundation - The Apache Way

A nonprofit providing a neutral home for open source projects, guided by community over code

The Apache Software Foundation (ASF) is a nonprofit that provides software for the public good. It is the steward of the Apache License 2.0, a permissive open source license used by many projects including Strands Agents, OpenSearch, and Apache Iceberg.

The Apache Way

Every Apache project follows four core principles that define how open source communities should operate:

Community over code

The people and culture matter more than the software. A healthy community produces better software than any single organization working alone.

Earned authority

Influence comes from contributions, not from job title, seniority, or employer. A volunteer who consistently fixes bugs, writes documentation, or mentors newcomers has the same voice as any senior engineer.

Open development

All decisions happen in public. Transparency is required in every Apache project.

Consensus decision making

Major decisions require community agreement. No single company can control the direction of an Apache project.

Popular Projects

Apache HTTP Server

Provides the open-source web server that powers a large share of all websites on the Internet. One of the oldest and most influential open-source projects ever created.

Apache Kafka

Created at LinkedIn and donated to the Apache Software Foundation. Acts as a distributed publish-subscribe system for real-time event streaming with high throughput and fault-tolerant processing.

Apache Spark

Originated at the University of California, Berkeley (AMP Lab) and later became an Apache project. Delivers an in-memory engine for large-scale data processing, machine-learning, and streaming analytics.

Apache Iceberg

Initiated at Netflix and contributed to Apache. Provides an open table format for petabyte-scale analytic datasets, enabling reliable schema evolution and hidden partitioning.

The Eclipse Foundation - Open Source Beyond Code

A nonprofit providing neutral, community-governed homes for open source projects worldwide

The Eclipse Foundation was originally created to govern the Eclipse IDE for Java developers, it has grown into a neutral home for projects spanning cloud infrastructure, edge computing, IoT, automotive software, and AI.

What the Eclipse Foundation Does

Neutral governance

Vendor neutral home for open source projects. No single company controls project direction. Decisions are made by the community of contributors, and competing companies can collaborate on shared technology without one party holding undue influence.

Intellectual property management

The foundation serves as the legal steward for trademarks, copyrights, and licensing across all hosted projects. **Vendor-**

Neutral collaboration

Companies that compete in the market can work together on shared open source infrastructure under the foundation's governance model. The structure provides a legally safe and neutral environment for cross company collaboration.

Notable Projects

Eclipse IDE

The original Java development environment that sparked the foundation. Still widely used by enterprise Java developers around the world. Supports many additional languages through plugins.

Jakarta EE

The enterprise Java platform, formerly Java EE. The Eclipse Foundation took stewardship from Oracle and continues its open, community-driven development.

Eclipse Mosquitto

An open source MQTT message broker widely used in IoT applications, from smart home devices to industrial sensors and connected vehicles.

SECTION 03

Contributing to Open Source

There is a role for everyone in open source

Why You Should Contribute to Open Source

Even if you never contribute a single line of code, understanding open source changes how you see the technology around you. If you do contribute, the impact compounds in ways that are hard to find anywhere else.

You already use it every day

Android phones, web browsers, programming languages, AI frameworks, and the servers powering websites you visit all run on open source. Understanding it means understanding how modern technology actually works.

You learn from how real systems are built

Open source lets you read production code written by engineers building systems at scale. This is not tutorial code or examples. It is the real thing, with a full history of decisions and changes available for anyone to study.

A public portfolio of real work

A merged pull request is public, permanent evidence of real work. It shows how you reason through problems, how you collaborate, and how you handle feedback. That kind of record is difficult to fake and hard to ignore.

You discover what the field is actually looks like

Contributing exposes the full range of roles in technology: engineering, documentation, security, developer advocacy, program management, community building. Many people find their direction by seeing the ecosystem up close.

Skills transfer everywhere

Open source skills are not platform-specific or employer-specific. Git workflows, code review, distributed collaboration, technical writing, and community communication apply everywhere.

Software is infrastructure

The systems that shape people's lives are increasingly built on software. When that software is open, it can be independently audited, studied, and challenged by anyone.

Code Contributions

When most people think of open source, they think of code



Bug Fixes

Find a broken feature? Fix it and submit a pull request. Great starting point for new contributors, even small fixes matter.



New Features

Propose and implement enhancements to a project.



Security Patches

Identifying and fixing vulnerabilities.



Performance

Making code faster, more memory-efficient, or more scalable.



Tests

Writing automated tests ensures software works correctly.



Code Reviews

Reviewing pull requests from other contributors. Catching bugs and giving feedback is extremely valuable.

Non-Code Contributions - Just as Important

Here's how to contribute without code:



Documentation

Writing or improving README files, wikis, tutorials, and API docs.



Design & UX

Creating logos, improving UI layouts, and producing design assets.



Community Support

Answering questions on GitHub Issues, Stack Overflow, Discord, or forums.



Translation

Translating software, documentation, and tutorials into other languages.



Outreach & Marketing

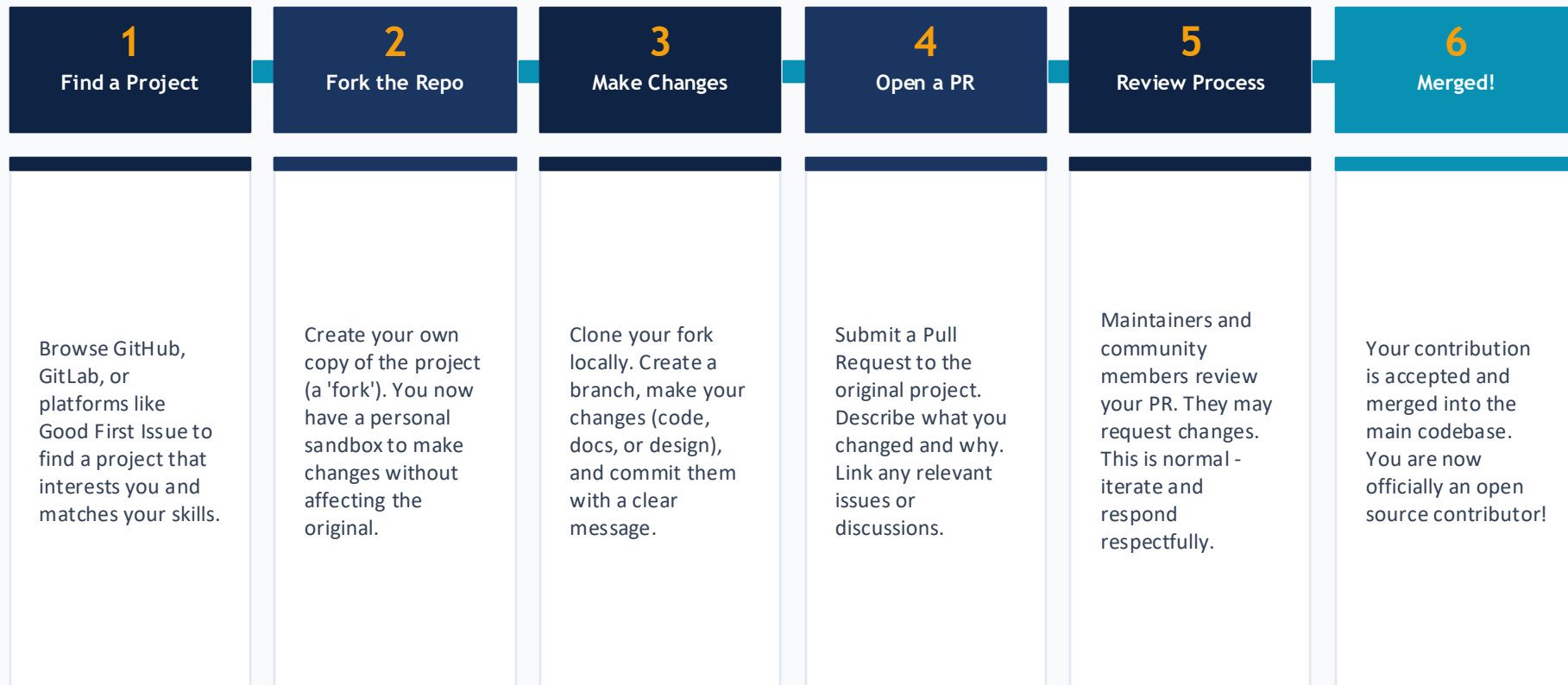
Writing blog posts, giving talks, or making videos about open source projects. Awareness drives adoption and attracts contributors.



Mentorship

Guiding new contributors through their first pull request.

How Contributions Work - The Pull Request Workflow



SECTION 04

Amazon's Open Source Contributions

How one of the world's largest tech companies gives back

Amazon Web Services Contributions

AWS is proud to support open source projects, foundations, and partners. We believe that open source is good for everyone and we are committed to bringing the value of open source to our customers.

OpenSearch

A search engine tool that helps you find information in large amounts of data, similar to how Google searches the web but designed for searching through a company's documents, logs, and databases.

Valkey

A high-performance, open source in-memory data store that helps systems load faster by storing frequently accessed information in memory, offering caching and real-time data processing capabilities.

Apache Iceberg

Apache Iceberg is a way to store big data tables on cloud storage that keeps a version history of every change, so you can query the data exactly as it looked yesterday, last week, or any time in the past

Strands

The Strands Agents SDK empowers developers to quickly build, manage, evaluate and deploy AI-powered agents.

MCP

MCP (Model Context Protocol) is a universal connector that gives AI assistants a single, consistent way to talk to many different apps and data sources.

Cedar

An open source authorization system that lets developers control who can access what in their applications by writing simple security policies, similar to setting permissions on your phone but for complex software systems.

SECTION 05

Open Source Licenses

Understanding the rules of the road

Common Open Source Licenses - What They Mean

A license defines what others can and cannot do with the code. Always check the license before using open source software in a project.

MIT License

Permissive

✓ Use, modify, distribute, sublicense, sell

⚠ Include the original license notice

Used by: React, Node.js, jQuery, .NET

Apache 2.0

Permissive

✓ Use, modify, distribute, patent rights granted

⚠ State changes made, include NOTICE file

Used by: Android, Kubernetes, TensorFlow

GPL v3

Copyleft

✓ Use, modify, distribute

⚠ Derivatives MUST also be GPL-licensed

Used by: Bash, GNU Coreutils

LGPL

Weak Copyleft

✓ Link from proprietary software

⚠ Modifications to the library must stay LGPL

Used by: FFmpeg, GTK

BSD 2/3-Clause

Permissive

✓ Use, modify, distribute

⚠ Keep copyright notice, no endorsement

Used by: FreeBSD, Go(Golang), OpenSSH, Valkey

GPL v2

Copyleft

✓ Use, modify, distribute

⚠ Derivatives MUST also be GPL-licensed

Used by: Linux kernel, WordPress (GPL v2 (or Later)), Git

SECTION 06

How to Get Started

Your roadmap to becoming an open source contributor

Your First Open Source Contribution - Step by Step

01

Learn Git Basics

Git is the version control system used by nearly all open source projects. Learn git clone, commit, push, pull, and branch. Free resources: <https://learn.github.com/skills>

02

Pick the Right Project

Start with something you USE. Check for a CONTRIBUTING.md file. Filter issues by 'good first issue' or 'beginner friendly' labels on GitHub.

03

Read Before You Write

Read the CONTRIBUTING.md, CODE_OF_CONDUCT.md, and existing issues before doing anything. Understand the project's goals, style, and communication norms.

04

Start Small

Fix a typo in documentation, improve an error message, or add a missing test. Small contributions build trust and help you learn the codebase incrementally.

05

Communicate

Comment on the issue you want to fix. Ask questions, be patient, respectful, and incorporate feedback.

Key Takeaways

1

Open source is software anyone can use, study, modify, and distribute. It powers the modern internet and most of the world's digital infrastructure.

2

The Linux Foundation is the world's largest open source organization, enabling neutral collaboration between competitors.

3

You don't need to code to contribute. Documentation, design, translation, and community support are just as valuable.

4

Amazon contributes to major projects like OpenSearch, Valkey, Apache Iceberg, Strands, MCP and Ceder

5

Getting started is easier than you think. Find a project you use, read its CONTRIBUTING.md, and start with something small. Everyone starts somewhere.

FAQ - Getting Started

Q

Do I need to be a good programmer to contribute to open source?

No. Many valuable contributions are non-code: documentation, translations, design, community support, and bug reporting. Even experienced developers start with small fixes. The key is willingness to learn and communicate respectfully.

Q

How do I find a beginner-friendly project?

Go to goodfirstissue.dev or filter GitHub issues by the label 'good first issue' or 'beginner friendly.' Pick a project you personally use. It's easier to contribute when you understand what the software is supposed to do.

Q

What if I make a mistake in a pull request?

That's completely normal and expected. Maintainers will review your PR and ask for changes. Simply update your code based on their feedback and push again. No PR is perfect on the first try, iteration is a part of the process.

Q

Is open source contribution good for my career?

Yes, significantly. It demonstrates real-world collaboration skills, builds a public portfolio of work, expands your network, and shows future employers your code in practice.

You're Ready to Contribute.

Open source is one of the most powerful forces in technology and it's powered entirely by people like you.

Start small. Be curious. Ask questions. Every expert was once a beginner, and every contribution, no matter how small makes the ecosystem better for everyone.

Good luck, and welcome to the community.

SPOTLIGHT

AWS Strands Agents

AWS's open source SDK for building AI agents

What Is Artificial Intelligence (AI)? (Before We Talk Strands)

Artificial Intelligence is the field of computer science focused on building systems that can perform tasks that normally require human intelligence - such as understanding language, recognizing images, writing code, and reasoning through problems.

Artificial Intelligence (AI)

The broad field - any technique making computers perform tasks that normally require human intelligence

Machine Learning (ML)

A subset of AI - systems that learn patterns from data without being explicitly programmed for every rule

Deep Learning

A subset of ML - uses multi-layered neural networks to learn highly complex patterns from large datasets

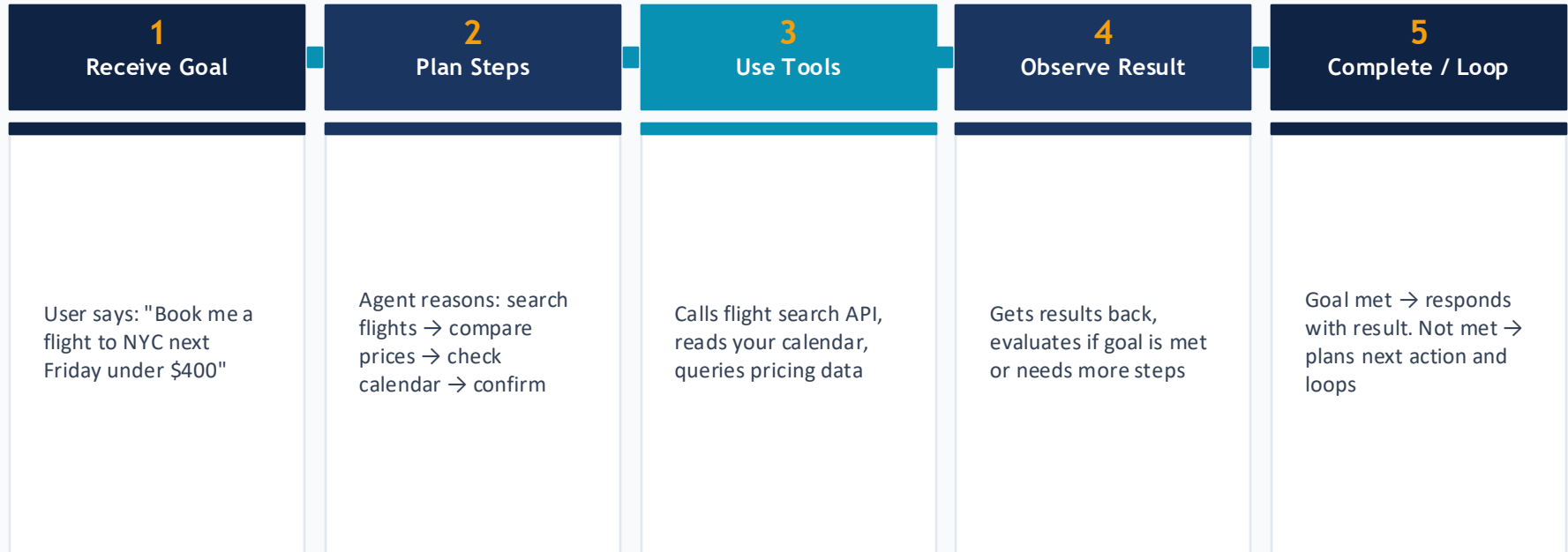
Large Language Models (LLMs)

A subset of Deep Learning - neural networks trained on massive text data to understand and generate human language. Examples: GPT-4, Claude, Llama 3, Gemini.

LLMs are the technology behind ChatGPT, Claude, Gemini, and most modern AI assistants

What Is an AI Agent? (Before We Talk Strands)

An AI agent is more than a chatbot. Where a chatbot responds to a single message, an agent can plan a sequence of steps, use tools (search the web, run code, query a database, call APIs), observe the results, and loop until a goal is accomplished - autonomously.



What Are Large Language Models (LLMs)?

A Large Language Model (LLM) is a type of AI trained on massive amounts of text data to understand, generate, and reason about human language. It powers chatbots, code assistants, search engines, and much more.

How LLMs Work

LLMs are trained by feeding them billions of sentences from books, websites, code, and articles. The model learns to predict the next word in a sequence, billions of times over until it understands grammar, facts, reasoning patterns, and context.

Once trained, you give the model a prompt (a question or instruction), and it generates a response one token (roughly one word) at a time, choosing the most likely continuation based on everything it learned.

This simple mechanism to predict the next word turns out to be powerful enough to write essays, debug code, answer complex questions, and hold conversations.

GPT-4o

By OpenAI

Closed

Claude 3.5

By Anthropic

Closed

Gemini 1.5

By Google

Closed

Llama 3

By Meta

Open Weight

Mistral 7B

By Mistral AI

Open Weight

Gemma 2

By Google

Open Weight

Teal = Open Weight | Grey = Closed (API only)

How AI Models Learn - Understanding Weights

you first need to understand how a model is built and what it is once training is complete.

What are weights?

A neural network is made of billions of connections, like a vast web of neurons inspired by the human brain. Every single connection has a number attached to it called a weight.

During training, the model is shown enormous amounts of data (text, images, code). It makes predictions, checks whether it was right or wrong, and then adjusts its weights slightly, getting better each time through a process called backpropagation.

After a period of time of training on thousands of GPUs, the process is complete. The final collection of all those numbers, all the weights is the model. It encodes everything the AI has learned.

1

Training Data

Trillions of words from books, code, websites

2

Neural Network

Billions of connections with adjustable weights

3

Training Loop

Model predicts → checks error → adjusts weights (repeat billions of times)

4

Weights File

Final numbers saved - this is the model

5

Inference

Load weights, give a prompt, get a response

What Is an Open Weight Model?

An open weight model is an AI model whose trained parameters (the 'weights') are made publicly available for anyone to download, run, and build upon - similar to how open source software shares code.

CLOSED AI MODEL



- You query the model via an API
- The weights live on the company's servers
- You cannot inspect or modify it
- Access can be revoked or priced anytime

VS

OPEN WEIGHT MODEL



- Download the weights to your machine
- Run entirely locally with no API needed
- Inspect, fine-tune, and modify freely
- No usage limits, no data sent to anyone

AWS Strands Agents SDK - What It Is

Released by AWS in May 2025, Strands Agents is a fully open source Python and TypeScript SDK for building AI agents. It takes a model-driven approach: instead of writing complex orchestration logic, you describe what you want the agent to do and give it tools. The model figures out the rest.

Model-Agnostic

Works with any AI model: Amazon Bedrock (Claude, Llama, Titan), Anthropic, OpenAI, Gemini, Ollama, LiteLLM, and more. Swap providers in one line of code. Not locked to AWS or any single provider.

Tool Use Built-In

Agents can be equipped with tools: Python functions, AWS services, web APIs, databases. Write a function, decorate it with `@tool`, and the agent calls it autonomously when needed. No orchestration code required.

MCP Support

Strands supports the Model Context Protocol, the open standard for connecting agents to external tool servers. Connect to any MCP server and the agent immediately gains access to its full set of capabilities.

Multi-Agent Built-In

Version 1.0 introduced Graph, Swarm, and Workflow patterns for multi-agent systems. One agent can orchestrate others. Includes the Agent-to-Agent (A2A) protocol.

Minimal Abstraction

A working agent requires very few lines of Python. Simple things stay simple. Complex architectures are still possible, but the SDK never forces unnecessary complexity on beginners or production teams.

Open Source - Apache 2.0 License

Built by AWS, licensed under the Apache License 2.0. This means you can use it commercially, modify it, and redistribute it freely.

Key Takeaways - AWS Strands Agents SDK

01

Agents take actions, chatbots just respond

The difference between ChatGPT and a Strands agent is tools. An agent can call APIs, read files, query databases, and take multi-step actions. A chatbot can only produce text.

03

Tools are what give agents power

http_request, filesystem MCP, Valkey tools, OpenSearch tools - each one expands what the agent can do. You compose capabilities by passing tools=[...]. The agent chooses which to use.

05

You already know enough to build something real

Phase 1 of the project is under 20 lines of Python.

02

The system prompt is your program

You do not write logic to control an agent. You write instructions in plain English. The agent reads them and decides what to do. Changing behavior means changing the prompt, not rewriting code.

04

Apache 2.0 means it is free for any use

Strands is open source under the Apache 2.0 license. Built by AWS. Use it commercially, modify it, contribute to it. The community owns the future of the SDK.

06

Where to go next

strandsagents.com/docs/examples - official examples including weather, memory, file operations, multi-agent workflows.
github.com/strands-agents - full source code, samples, and community discussions.

Key Takeaways - Open Weight Models

Open weight is NOT open source AI

Most models called 'open source' like Llama, and Gemma are actually open weight. They release the model parameters but not the training data or code.

Fine-tuning is the real power

You can take a general model and train it further on your own data to create a specialist. A hospital can fine-tune for medical language. A law firm for legal documents. This level of customization is impossible with closed APIs.

How to run one today

Ollama (ollama.com) - run Llama or Gemma on your laptop with one terminal command. Hugging Face (huggingface.co) - browse and run models in your browser. No setup, no API key.

SPOTLIGHT

Valkey

Valkey is an open source in-memory data store supported by AWS.

First: What Is an In-Memory Cache?

Before we explain Valkey, you need to understand the concept it is built on. A cache is a place to store things temporarily so you can get them back faster next time. An in-memory cache does this in RAM, which is thousands of times faster than reading from a hard drive or making a network call.

The Hard Drive / Database

Where your data permanently lives. Reliable and large, but slow to read.

The Network Call / API

Fetching data from another server across the internet. Even slower. Can fail. Costs money per request.

The In-Memory Cache

A copy of your most-used data stored directly in RAM. Retrieval takes microseconds, before even thinking about the database.

Your Application

Checks the cache first. If the data is there (a cache 'hit'), instant response. If not (a cache 'miss'), fetch from the database and store it in the cache for next time.

What Is Valkey?

Open source in-memory data store governed by the Linux Foundation

Valkey is a open source, in-memory key-value data store. Think of it as a notebook that never gets lost. You write something down, your program can read it back instantly, even after it restarts. It stores data in RAM so retrieval is near-instant.

What 'key-value' means

Every piece of data has a label (the key) and a value. Key: 'user_city'. Value: 'Seattle'. You retrieve anything by its key in under a millisecond. Like a dictionary you can look up at any time.

Why RAM makes it fast

Data lives in memory (RAM). Reading from RAM takes microseconds. Reading from disk takes milliseconds. For frequent lookups, that difference adds up to a dramatically faster application.

Persistence across restarts

Despite living in RAM, Valkey can write snapshots to disk. When your program restarts, it reads those snapshots back. Your agent's preferences and history survive shutdowns.

SETEX: automatic expiry

SETEX stores a value with a timer. After the time runs out, Valkey deletes it automatically. Perfect for caching news headlines for one hour or weather data for two hours without any cleanup code.

Where Valkey came from

In 2024, Redis changed its open source license. The community, including AWS and Google, forked the project to create Valkey under the BSD license. It is now governed by the Linux Foundation.

Three commands cover most use cases

SET stores a value. GET retrieves it. SETEX stores with an expiry. These three commands handle the vast majority of real use cases. Valkey is powerful precisely because its core API is this simple.

Key Takeaways - Valkey

01

An in-memory key-value database

Valkey stores data as key value pairs and is optimized for fast lookups rather than relational or SQL style queries

02

Speed comes from living in RAM

Valkey stores data in memory (RAM), enabling fast reads and writes compared to disk based databases. Best used for frequently accessed data that needs to be instant.

03

Persistence means data survives restarts

Despite being in-memory, Valkey can write snapshots to disk. Your agent's preferences and cached data survive program restarts. The next time you run your script, everything is already there waiting.

04

SETEX is the most useful command you will learn

SET stores a value permanently. SETEX stores with an expiry time. Use SETEX to cache news headlines for one hour, weather for two hours, or any data that should refresh automatically without manual management.

05

BSD licensed and community governed

Valkey was forked from Redis in 2024 when Redis changed its license. AWS, Google, and others co-founded the project under the Linux Foundation with a BSD-3 Clause license. It is free for any use.

06

Four commands cover nearly everything

SET stores a value. GET retrieves it. SETEX stores with an expiry. DEL removes a key. These four commands handle the vast majority of real use cases. Valkey is powerful precisely because its core API is simple.

Thank You.

AWS Open Source

aws.amazon.com/opensource

All AWS open source projects and contributions

Strands Agents SDK

strandsagents.com

Official docs, examples, and community

Valkey

valkey.io

Official docs, and community